MATH 128a: Homework #5

Instructor: F. Weber

Due by: **Wednesday, October 8, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Chapter 1

(a) Show that the sequence $a_n = e^{-\frac{1}{n^2}}$ satisfies $a_n = 1 + O(n^{-2})$ as $n \to \infty$.

Solution. We write out the Taylor series for e^x

$$e^x = 1 + O(x)$$

and substitute our specific value of x

$$e^{-1/n^2} = 1 + O(1/n^2)).$$

We note that $O(1/n^4)$ sits inside $O(1/n^2)$. Also $1/n^2$ is $O(1/n^2)$. Thus

$$e^{-1/n^2} = 1 + O(1/n^2).$$

(b) Show that the equation $0 = x^2 - 4x^3 + 1$ has at least one solution in the interval $[0, 1]$.

Solution. Consider $f(x) = x^2 - 4x^3 + 1$, which is a continuous function because it is a polynomial. Observe that $f(0) = 1$ and $f(1) = -2$. Thus, the **intermediate value theorem** tells us there exists a $c \in (0, 1)$ such that $f(c) = 0$.

(c) Find the extrema of the function $f(x) = x/(1 + x)$ on the interval $[0, 4]$.

$$f'(x) = \frac{1}{(1+x)} - \frac{x}{(1+x)^2} = \frac{1}{(1+x)^2}$$

so $f'(x)$ has no zeros on $[0, 4]$. Thus we need only compare $f(0) = 0$ and $f(4) = 4/5$. We conclude that $f(0) = 0$ is the minimum and $f(4) = 4/5$ is the maximum.

(d) Do problem 10(8) in Section 1.1 of the textbook.

We have

$$f = \sqrt{x+1}, \quad f' = \frac{1}{2\sqrt{x+1}}, \quad f'' = \frac{-1}{4(x+1)^{3/2}}, \quad f''' = \frac{3}{8(x+1)^{5/2}}$$

so the Taylor series is

$$1 + \frac{1}{2}x + \frac{-1}{4}\frac{x^2}{2!} + \frac{3}{8}\frac{x^3}{3!}$$

x	-0.5	0.-0.25	0.25	0.5
P_3	0.7109	0.8862	1.1182	1.2266
Error	.0038	.0002	.0001	.0018

(e) Do problem 18 of Section 1.1 of the textbook.

Solution. We compute

$$f_n(x) = \frac{n!}{(1-x)^n}$$

so

$$P_n(x) = \sum_{k=0}^{n} x^k$$

and

$$R_n(x) = \frac{x^{n+1}}{(1+\zeta(x))^{n+1}}$$

which is maximized when $x = 1/2$ and $\zeta(x) = 0$, i.e.

$$|R_n(x) \leq \frac{1}{2^{n+1}}\frac{1}{(1+0)^{n+1}} = 1/2^{n+1}$$

so we need

$$1/2^{n+1} \leq 10^6 \iff n \geq 6\log_2(10) - 1 \approx 18.9.$$

Thus $n = 19$ will do.

(f) Do problem 1 b) and c) of Section 1.2 of the textbook.

Solution.

b. Absolute error: $|\pi - 3.1416| \approx 7.35 \times 10^{-6}$. Relative error: $|\pi - 3.1416|/\pi \approx 2.34 \times 10^{-6}$.

c. Absolute error: $|e - 2.718| \approx 2.82 \times 10^{-4}$. Relative error: $|e - 2.718|/e \approx 1.04 \times 10^{-4}$.

(g) Solve problem 20b) (16b) of Section 1.2 of the textbook.

(h) This number is

$$(-1)2^{1034-1023}(1 + 1/2 + 1/16 + 1/128 + 1/256) = -3224$$

The next largest number is

$$-3224 + 2^{11-52}$$

and the next smallest number is

$$-3224 - 2^{11-52}.$$

This content is protected and may not be shared, uploaded, or distributed.

Problem 2: Bisection method

(a) What is going wrong in the following code for the bisection method? (aka time to debug!)

Listing 1: Code for bisection method

```
function [p,nit] = bisection(f,a,b,tol)

if nargin < 4
    tol = 10^(-6);
    if nargin < 2
        a=0;
        b=1;
        if nargin < 1
            f = @(x)(x^2-0.5);
        end
    end
end


% solve f(p)=0 using the bisection method, stop when we find a point that
% is closer than tol to p
% nit number of iterations needed


for k = 1:10000
    p = (a+b)/2;
    nit = k;
    if abs(p-a)< tol || f(p)==0, break; end
    if f(a)*f(b)>0
        a = p;
    else
        b = p;
    end
end
```

Solution. The statement `if f(a)*f(b)>0` should be changed to `if f(a)* f(p)>0`.

(b) Implement a different stopping criterion than the one used in this code and compare to the original one.

Solution. We could try the stopping criterion `abs(f(p))<tol` instead. In this case, we could guarantee that we have found a point p such that $f(p)$ is indeed small, but the point may be very far from a true zero p^*. On the other hand, with the current stopping criterion, we guarantee that we have fount a point p that is very close to a true zero p^*, but it might be that $f(p)$ is still very large.

Problem 3: Fixed point iteration

Solve problem 3 of Section 2.2 of the textbook.

This content is protected and may not be shared, uploaded, or distributed.

a. $0 = x^3 - 2x + 1 \implies 2x = x^3 + 1 \implies x = \frac{1}{2}(x^3 + 1)$.

$$p_1 = 9/16, \quad p_2 \approx 0.5890, \quad p_3 \approx 0.6022, \quad p_4 \approx 0.6092$$

b. $0 = x^3 - 2x + 1 \implies 0 = x - 2/x + 1/x^2 \implies x = 2/x - 1/x^2$.

$$p_1 = 0$$

and then p_2 is undefined because it involves division by zero.

c. $0 = x^3 - 2x + 1 \implies x^2 = 2 - 1/x \implies x = \sqrt{2 - 1/x}$

$$p_1 = 0$$

and then p_2 is undefined because it involves division by zero.

d. $0 = x^3 - 2x + 1 \implies x^3 = 2x - 1 \implies x = -\sqrt[3]{1 - 2x}$.

$$p_1 = 0, \quad p_2 = -1, \quad p_3 = -\sqrt[3]{3} \approx -1.4422, p_4 \approx -1.5720$$

It seems that methods (a.) and (d.) are appropriate because the distance between consecutive iterations gets smaller each time.

This content is protected and may not be shared, uploaded, or distributed.